Decision and Risk Analysis

Lecture 2: **Bayes' Rule**

DAEN 427/ISEN 427/ISEN 627

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Computer Science > Social and Information Networks

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Fake News Detection on Social Media: A Data Mining Perspective

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Social media for news consumption is a double-edged sword. On the one hand, its low cost, easy access, and rapid dissemination of information lead people to seek out and consume news from social media. On the other hand, it enables the wide spread of "fake news", i.e., low quality news with intentionally false information. The extensive spread of fake news has the potential for extremely negative impacts on individuals and society. Therefore, fake news detection on social media has recently become an emerging research that is attracting tremendous attention. Fake news detection on social media presents unique characteristics and challenges that make existing detection algorithms from traditional news media ineffective or not applicable. First, fake news is intentionally written to mislead readers to believe false information, which makes it difficult and nontrivial to detect based on news content; therefore, we need to include auxiliary information, such as user social engagements on social media, to help make a determination. Second, exploiting this auxiliary information is challenging in and of itself as users' social engagements with fake news produce data that is big, incomplete, unstructured, and noisy. Because the issue of fake news detection on social media is both challenging and relevant, we conducted this survey to further facilitate research on the problem. In this survey, we present a comprehensive review of detecting fake news on social media, including fake news characterizations on psychology and social theories, existing algorithms from a data mining perspective, evaluation metrics and representative datasets. We also discuss related research areas, open problems, and future research directions for fake news detection on social media.

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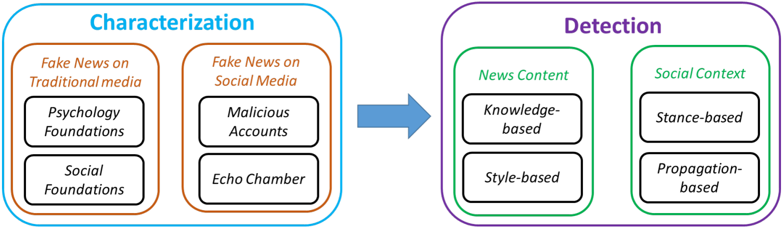
Framework



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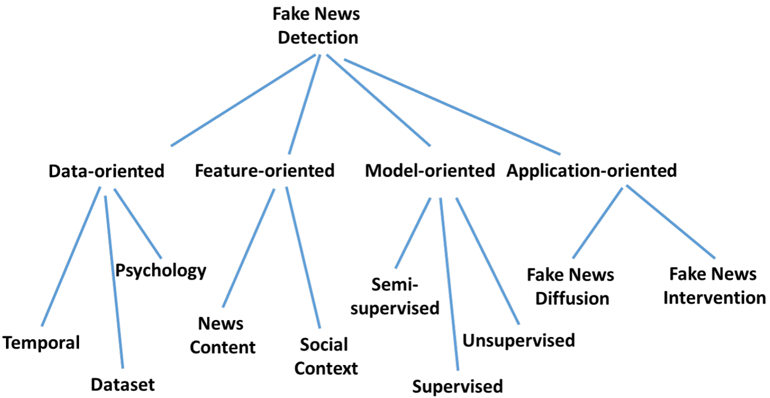
Future Directions



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Building a Bayesian Model for Events

Before reading anything,
we can use prior knowledge from historical data.

Valid Probability Model

A probability model must satisfy:

Building a Bayesian Model for Events

Before reading anything,
we can use prior knowledge from historical data.

Valid Probability Model

A probability model must satisfy:

1. Cover all possible events.
2. Assign probabilities to each event.
3. Probabilities sum to one.

We're Using R Instead of Python

Python

- "Can do everything" ... installs n packages, breaks after update
- Stack Overflow becomes your TA

R $\rightarrow$ `https://cran.r-project.org`

- Literally for statistics; `lm(y ~ x)` and move on with your life
- Excellent, open-source, keeps us focused on data analysis,
  ... and I enjoy sleeping at night!

What I Need From You: learn concepts, analyze data, pass the class!
What I Do Not Need: a 2-hour debate about virtual environments.

Dataset

```r
# Load packages
library(bayesrules)
library(tidyverse)
library(janitor)

# Import article data
data(fake_news)

fake_news %>%
  tabyl(type) %>%
  adorn_totals("row")
#  type   n percent
#  fake  60     0.4
#  real  90     0.6
# Total 150     1.0
```

Prior Probability Model

- 150 news articles:
  60 fake, 90 real
- $P(\text{Fake}) = \frac{60}{150} = 0.40$.
  Before looking at the headline, the prior probability that an article is fake is 40%.

Prior Belief: there is a 40% chance an article is fake before observing any evidence.

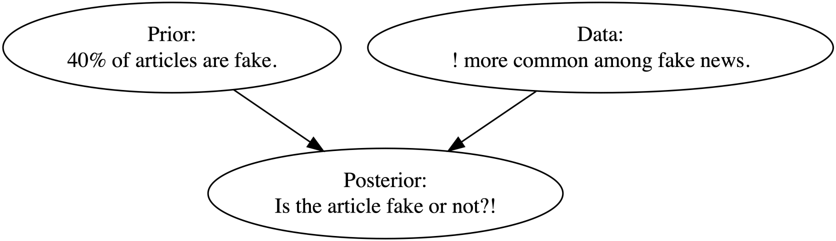
$$P(B) = 0.40 \quad \leftrightarrow \quad P(B^c) = 0.60$$

where B: article is **fake**, B^c: article is **real**.

$$P(B) + P(B^c) = 0.40 + 0.60 = 1$$

Event	B	B^c	Total
Probability	0.4	0.6	1

Bayesian reasoning to estimate whether a news article is fake:



DAG: Directed Acyclic Graph

$$A \longrightarrow B$$

$A =$ exclamation points "!" $B =$ fake status "🥸"

- A **DAG** shows how variables are connected.
- The arrow means A may be associate with B.
- **Acyclic** means the arrows do not loop back.

DAGs and Conditional Probability

- The DAG suggests that the chance of B may depend on A.

$$P(B \mid A)$$

In words: the probability of article is fake **given**
the exclamation point in the title.

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$$P(B \mid A) \neq P(B)$$

DAGs and Conditional Probability

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$$P(B \mid A)$$

In words: the probability of article is fake **given** the exclamation point in the title.

- If A changes the probability of B, then the events are **dependent**.

$$P(B \mid A) \neq P(B)$$

- If knowing A does not change the probability of B, then they are **independent**.

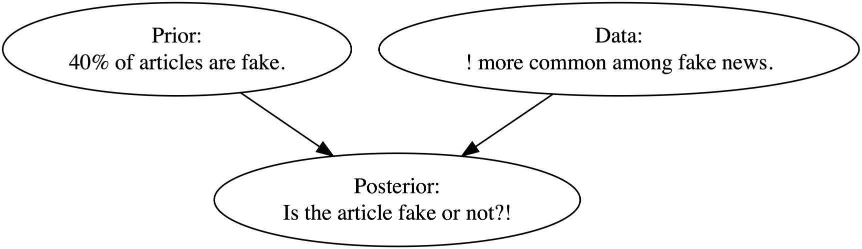
$$P(B \mid A) = P(B)$$

```r
# Transposed version of head()
# ... makes possible to see all columns in a data frame
glimpse(fake_news)
```

Observed headline pattern:

```r
# Tabulate exclamation usage and article type
fake_news %>%
  tabyl(title_has_excl, type) %>%
  adorn_totals("row")
# title_has_excl fake real
#          FALSE   44   88
#           TRUE   16    2
#          Total   60   90
```

Exclamation Points as Evidence



Title Has "!"	Fake	Real	Total
Yes	16	2	18
No	44	88	132
Total	60	90	150

Conditional Probability & Likelihood

Title Has "!"	**Fake**	**Real**	**Total**
Yes	16	2	18
No	44	88	132
Total	60	90	150

$$P(! \mid \text{Fake}) = \frac{16}{60} \approx 0.267,\ P(! \mid \text{Real}) = \frac{2}{90} \approx 0.022$$

An exclamation point increases the probability that an article is fake.

Conditional Probability & Likelihood

$$B : \text{article is fake}$$
$$B^c : \text{article is real}$$

$$A : \text{article uses exclamation points}$$
$$A^c : \text{article does not use exclamation points}$$

From the data:

$$P(A \mid B) = 0.267 \qquad P(A \mid B^c) = 0.022$$

Fake news articles are **much more likely** to use exclamation points than real news articles.

Conditional vs Unconditional Probability

- **Unconditional probability** is the chance something happens without extra information.

$$P(A)$$

- **Conditional probability** is the chance something happens given that something else is already known.

$$P(A \mid B)$$

- Comparing $P(A)$ and $P(A \mid B)$ tells us whether knowing B changes how likely A is.

Independent Events

- Two events are **independent** if knowing one happens does not change the probability of the other.
- *Example*: rolling a die and flipping a coin are independent, because the die result **does not** affect the coin flip.
- In symbols:

$$P(A \mid B) = P(A)$$

From Probability to Likelihood

- Earlier, we computed:

$$P(A \mid B) = 0.2667 \qquad P(A \mid B^c) = 0.0222$$

- Fake articles are much more likely to use exclamation points.
- Suppose we already observed an article with exclamations.

  Now the question becomes:

  Is the article more likely fake or real?

What is Likelihood?

- **Probability** starts with a known cause and predicts data.

$$P(A \mid B)$$

- **Likelihood** starts with observed data and compares possible causes.

$$\mathcal{L}(B \mid A) = P(A \mid B)$$

- In our example:
    - A: article uses exclamation points
    - B: article is fake
    - Since

    $$P(A \mid B) > P(A \mid B^c)$$

    exclamation provide evidence that the article may be fake.

Probability	**Likelihood**
Known condition	Known data
Predicts outcomes	Compares explanations
$P(A \mid B)$	$\mathcal{L}(B \mid A)$

- The distinction is **subtle**, especially since people use the terms "likelihood" and "probability" interchangeably in casual conversation.
- Likelihood **helps us decide** which explanation best fits the observed data.

- The likelihood function **is not** a probability function, but rather provides a framework to compare the relative compatibility of our exclamation point data with B and B^c.

Event	B	B^c	**Total**
Prior probability	0.4	0.6	1
Likelihood $(\cdot \mid A)$	0.2667	0.0232	0.2889

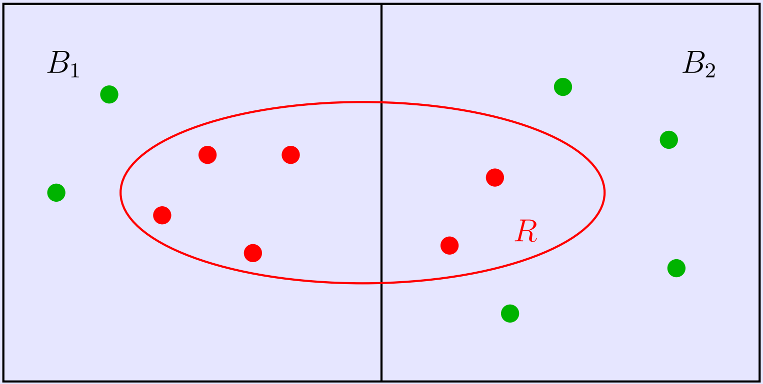
Quiz

Two boxes are available:

- Box 1 contains 4 red and 2 green balls.
- Box 2 contains 2 red and 4 green balls.
- The probability of choosing either box is equal:

$$P(E_1) = P(E_2) = \frac{1}{2}.$$

- A box is selected at random, and one ball is drawn.
- Study the **likelihood** of observing a red ball under each box.


B_1
B_2
R

Answer: A red ball is more consistent with Box 1 than with Box 2.

The likelihoods are: $P(R \mid B_1) = \frac{4}{6} = \frac{2}{3}$ and $P(R \mid B_2) = \frac{2}{6} = \frac{1}{3}$.

So, if Box 1 had been selected, observing a red ball would be more likely.

$$P(R \mid B_1) > P(R \mid B_2)$$

Therefore, the red ball provides stronger evidence for Box 1 than for Box 2.

$$\frac{P(R \mid B_1)}{P(R \mid B_2)} = \frac{\frac{2}{3}}{\frac{1}{3}} = 2$$

The observed red ball is **twice** as likely under Box 1 as under Box 2. Since the boxes were equally likely before the draw, the red ball favors Box 1.

Recap

$$A = \text{article uses exclamation points}$$
$$A^c = \text{article does not use exclamation points}$$
$$B = \text{article is fake news}$$
$$B^c = \text{article is real news}$$

From the prior model:

$$P(B) = 0.4 \qquad P(B^c) = 0.6$$

So, 40% of articles are fake and 60% are real.

Normalizing Constants

- The marginal probability of observing exclamation points across all articles is $P(A)$.
- This is called the **normalizing constant**.

Normalizing Constants

- The marginal probability of observing exclamation points across all articles is $P(A)$.
- This is called the **normalizing constant**.
- To calculate it, we split articles into two groups:
$$P(A) = P(A \cap B) + P(A \cap B^c)$$
- That is:
Exclamations $=$ fake with exclamations $+$ real with exclamations

Joint Probability Table

	B	B^c	Total
A	$P(A \cap B)$	$P(A \cap B^c)$	$P(A)$
A^c	$P(A^c \cap B)$	$P(A^c \cap B^c)$	$P(A^c)$
Total	0.4	0.6	1

- Each inside cell is a **joint probability**.
- For example: $P(A \cap B)$ means the probability that an article is fake and uses exclamation points.

Fake Articles **with** Exclamation Points

- To determine the probabilities of these joint events, note that 40% of articles are fake and 26.67% of fake articles use exclamation points, $P(B) = 0.4$ and $P(A \mid B) = 0.2667$.
- The joint probability is $P(A \cap B) = P(A \mid B)\, P(B) = 0.2667 \cdot 0.4 = 0.1067$
- So, 10.67% of all articles are fake and use exclamation points.

	B	B^c	Total
A	0.4 $\cdot$16/60 = **0.107**	$P(A \cap B^c)$	32/150 = 0.12
A^c	$P(A^c \cap B)$	$P(A^c \cap B^c)$	118/150 = 0.78
Total	**0.4**	0.6	1

Fake Articles **without** Exclamation Points

- Since 26.67% of fake articles use exclamation points, $1 - 0.2667 = 73.33\%$ do not use exclamation points.
- Then, $P(A^c \cap B) = P(A^c \mid B)\, P(B) = 0.7333 \cdot 0.4 = 0.2933$
- So, 29.33% of all articles are fake and do not use exclamation.

Checking the Fake Article Total

The fake-news column is split into two parts:

$$P(A \cap B) \quad \text{and} \quad P(A^c \cap B)$$

Therefore,

$$P(B) = P(A \cap B) + P(A^c \cap B)$$

$$P(B) = 0.1067 + 0.2933$$

$$P(B) = 0.4$$

This matches the prior probability that an article is fake.

Real Articles

From the likelihood function: $P(A \mid B^c) = 0.0222$

So only 2.22% of real articles use exclamation points.

The joint probability is

$$P(A \cap B^c) = P(A \mid B^c)P(B^c)$$

$$P(A \cap B^c) = 0.0222 \cdot 0.6$$

$$P(A \cap B^c) = 0.0133$$

So, 1.33% of all articles are real and use exclamation points.

Real Articles without Exclamation Points

Since

$$P(A \mid B^c) = 0.0222$$

we have

$$P(A^c \mid B^c) = 1 - 0.0222$$
$$P(A^c \mid B^c) = 0.9778$$

Then

$$P(A^c \cap B^c) = P(A^c \mid B^c)\, P(B^c)$$
$$P(A^c \cap B^c) = 0.9778 \cdot 0.6$$
$$P(A^c \cap B^c) = 0.5867$$

Completed Joint Probability Table

	B	B^c	Total
A	0.107	0.013	0.12
A^c	0.293	0.587	0.88
Total	0.4	0.6	1

The row total for A gives the **normalizing constant**:

$$P(A) = 0.107 + 0.013 = \boxed{0.12}$$

So, 12% of all articles use exclamation points.

Joint Probability

The calculations use the relationship

$$P(A \cap B) = P(A \mid B)P(B)$$

This connects: joint probability

with: conditional probability $\times$ prior probability

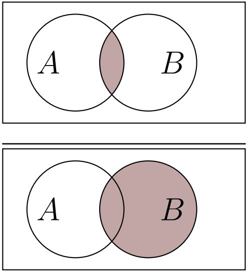
Rearranging gives the conditional probability formula:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

PS. If A and B are **independent**, then: $P(A \cap B) = P(A)\,P(B)$

Representation of Conditional Probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} =$$



Conditional Probability

Conditional probability answers the question:

> *"What is the probability of A given that B has occurred?"*

The definition is:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \qquad \text{where } P(B) \neq 0$$

- Start with the probability that both A and B occur
- Restrict attention only to situations where B occurs
- Measure how often A also occurs

Example: $P(\text{Rain} \mid \text{Cloudy}) = \frac{P(\text{Rain and Cloudy})}{P(\text{Cloudy})}$

Law of Total Probability

Suppose event A can happen in two distinct ways:

$$A \cap B \qquad \text{or} \qquad A \cap B^c$$

Since B and B^c cover all possibilities, we can write:

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

Using the joint probability rule:

$$P(A \cap B) = P(A \mid B)P(B)$$

we get the **Law of Total Probability**:

$$P(A) = P(A \mid B)P(B) + P(A \mid B^c)P(B^c)$$

Exclamation Points in News Articles

$$A = \text{article uses an exclamation point}$$
$$B = \text{article is fake news}$$

$$P(A \mid B) = 0.2667, \qquad P(B) = 0.4$$
$$P(A \mid B^c) = 0.0222, \qquad P(B^c) = 0.6$$

Then:

$$P(A) = 0.2667\,(0.4) + 0.0222\,(0.6)$$
$$P(A) = 0.1067 + 0.0133 = 0.12$$

So, about **12% of all articles** use exclamation points.

$$P(\text{Fake } |!) = \frac{P(\text{Fake})\, P(! \mid \text{Fake})}{P(!)}$$

Calculating Joint and Conditional Probabilities

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-
-

Posterior Probability Model via Bayes' Rule

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Posterior Simulation

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Building a Bayesian Model for Random Variables

Prior Probability Model

Discrete Probability Model

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-

The Binomial Data Model

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-
-

Conditional Probability Model of Data Y

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-
-

The Binomial Model

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-
-

The Binomial Likelihood Function

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-
-

Probability Mass Functions vs Likelihood Functions

Normalizing Constant

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-
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Posterior Probability Model

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Bayes' Rule for Variables

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-
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Posterior Shortcut

-
-
-

Proportionality

-
-
-

Posterior Simulation

Quiz 1

A medical test correctly detects a disease 95% of the time when the disease is present. This value represents:

a. Prior probability

b. Posterior probability

c. False positive rate

d. Likelihood

Answer: d. Likelihood

The statement describes: $P(\text{Test positive} \mid \text{Disease present}) = 0.95$

This is the probability of observing the test result given that the disease is present, which represents the likelihood (also called **sensitivity** or **true positive rate**).

- **Prior**: Probability of disease before observing the test result.
- **Posterior**: Probability of disease after observing the test result.
- **False positive rate**: Probability that the test is positive when the disease is absent.

Quiz 2

A university uses an AI system to detect plagiarism.

- 5% of assignments are actually plagiarized.
- The AI flags plagiarized assignments 92% of the time.
- The AI incorrectly flags honest assignments 8% of the time.

If an assignment is flagged, what is the probability it was actually plagiarized? Briefly discuss whether the AI system seems reliable.

Answer:

$$P(P \mid F) = \frac{0.92 \cdot 0.05}{0.92 \cdot 0.05 + 0.08 \cdot 0.95} \approx 0.377$$

Approximately 37.7% chance that it is actually plagiarism.
The system catches most plagiarism cases, but because plagiarism is rare, many flagged assignments are false positives.

Quiz 3

Let's suppose that you are an air traveler along with 99 other passengers from your flight, each of whom, like you, checked one item of luggage.

A recording piped through the speakers reminds you that: *"Many bags look alike. Please check your bag carefully before exiting the terminal."*

Indeed, your bag is one of the most popular models on the market, a medium-sized black rectangular case **used by 5% of all travelers**.

Given your distance from the luggage chute, you can discern only the shape, size, and color of the bags that emerge, not any detailed markings on the bags.

Quiz 3, cont.

a. Let's suppose that the **first bag** from your flight to enter the luggage carousel indeed has the same shape, size, and color as yours. Is it yours? Explain (i.e., calculate the posterior probability that the 1st bag is yours).

b. Now suppose that we continue to wait for our bag to appear, failing to see it among the first 98 bags to enter the carousel (i.e., only 2 bags are left to come off the airplane). Calculate the posterior probability that the **99th** bag, which also matches ours in shape, size, and color, is our own.

Answer: a. Let Y = first bag is yours, and M = first bag matches.

$$P(Y) = \frac{1}{100}$$
$$\mathcal{L}(Y^c \mid M) = P(M \mid Y^c) = 0.05$$
$$\mathcal{L}(Y \mid M) = P(M \mid Y) = 1$$

$$P(Y \mid M) = \frac{P(M \mid Y)P(Y)}{P(M)} = \frac{P(M \mid Y)P(Y)}{P(M \mid Y)P(Y) + P(M \mid Y^c)P(Y^c)} =$$
$$= \frac{1 \cdot 0.01}{1 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.01}{0.0595} \approx 0.168$$

So even though the first bag matches, it is probably not yours.

Answer: b.

$$P(Y \mid M) = \frac{1 \cdot \frac{1}{\text{total of bags remaining}}}{1 \cdot \frac{1}{\text{total of bags remaining}} + 0.05 \cdot \frac{\text{total of bags remaining} - 1}{\text{total of bags remaining}}}$$

(if 2 bags remain)

$$= \frac{1 \cdot \frac{1}{2}}{1 \cdot \frac{1}{2} + 0.05 \cdot \frac{1}{2}} = \frac{1}{1 + 0.05} \approx 0.9524$$

There is a 95% chance that the 99th bag is yours, given that it looks like yours.

Or, it is **19 times more likely** to be yours than not yours $^{(0.95/0.05=19)}$.

Homework

- Submit your answers before the next class.
- You may discuss with peers but submissions must be individual.
- Responses may be shared anonymously during class discussion!

Question 1

Define the following events for a resident of a fictional town as:

- $A =$ drives 10 miles per hour above the speed limit,
- $B =$ gets a speeding ticket,
- $C =$ took statistics at the local college,
- $D =$ has used R,
- $E =$ likes the music of Prince, and
- $F =$ is a Texan.

Several facts about these events are listed next.

Question 1, cont.

Specify each of these facts using probability notation, paying special attention to whether it’s a marginal, conditional, or joint probability.

- 73% of people that drive 10 miles per hour above the speed limit get a speeding ticket.
- 20% of residents drive 10 miles per hour above the speed limit.
- 15% of residents have used R.
- 91% of statistics students at the local college have used R.
- 38% of residents are Texans that like the music of Prince.
- 95% of the Texans like the music of Prince.

Question 2

Edward is trying to prove to Bella that vampires exist. Bella thinks there is a 0.05 probability that vampires exist. She also believes that the probability that someone can sparkle like a diamond if vampires exist is 0.7, and the probability that someone can sparkle like a diamond if vampires don't exist is 0.03. Edward then goes into a meadow and shows Bella that he can sparkle like a diamond. Given that Edward sparkled like a diamond, what is the probability that vampires exist?

Question 3

The probability that Gabby will like a restaurant is 0.7. Among the restaurants that she likes, 20% have five stars on Yelp, 50% have four stars, and 30% have fewer than four stars. What other information do we need if we want to find the posterior probability that Gabby likes a restaurant given that it has fewer than four stars on Yelp?

Question 4

Robin applies for six equally competitive data science internships. He has the following prior model for his chances of getting into any given internship, π:

π	0.3	0.4	0.5	Total
$f(\pi)$	0.25	0.60	0.15	1

(a) Let Y be the number of internship offers that Robin gets. Specify the model for the dependence of Y on π and the corresponding pmf, $f(y \mid \pi)$.

(b) Robin got some pretty amazing news. He was offered four of the six internships! How likely would this be if $\pi = 0.3$?

(c) Construct the posterior model of π in light of Robin's internship news.

Question 5

Whether you like it or not, cats have taken over the internet. Joining the craze, Michael has written an algorithm to detect cat images. It correctly identifies 80% of cat images as cats, but falsely identifies 50% of non-cat images as cats. Michael tests her algorithm with a new set of images, 8% of which are cats. What's the probability that an image is actually a cat if the algorithm identifies it as a cat? Answer this question by simulating data for 10,000 images.